



PH 142: Week 5 Review



Overview

- ◎ T-tests
 - One-sample
 - Two-sample
 - Paired sample
- ◎ ANOVA
 - Comparison of means of more than two groups
 - Tukey HSD
- ◎ Non-parametric tests
 - Wilcoxon Rank Sum
 - Wilcoxon Sign-Rank
 - Kruskal-Wallis
- ◎ Inference for regression



T-tests

One-sample t-test

- ◎ Goal: Test whether the mean of the sample is different than the mean of the population
- ◎ Conditions to meet:
 - Data is an SRS from a much larger population
 - Observations follow a normal distribution
 - Population standard deviation (SD) is **NOT** known
- ◎ Hypotheses:

$$H_0 : \mu = \mu_0$$

$$H_A : \mu \neq \mu_0, \text{ or } \mu < \mu_0, \text{ or } \mu > \mu_0$$

- ◎ Follows a t-distribution with $n-1$ degrees of freedom
- ◎ Test statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Two-sample t-test

Goal: Test if two population means are different

- Conditions to meet:
 - Samples are independent and came from a SRS (measure the same quantitative **continuous** variable for both populations)
 - Both populations follow a normal distribution (population means and SDs are unknown)
 - Hypotheses:

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_A : \mu_1 - \mu_2 \neq 0, \text{ or } \mu_1 - \mu_2 < 0, \text{ or } \mu_1 - \mu_2 > 0$$

Two-sample t-test cont.

- Sampling distribution for the difference between the two groups follows a t-distribution with degrees of freedom (use R to calculate):

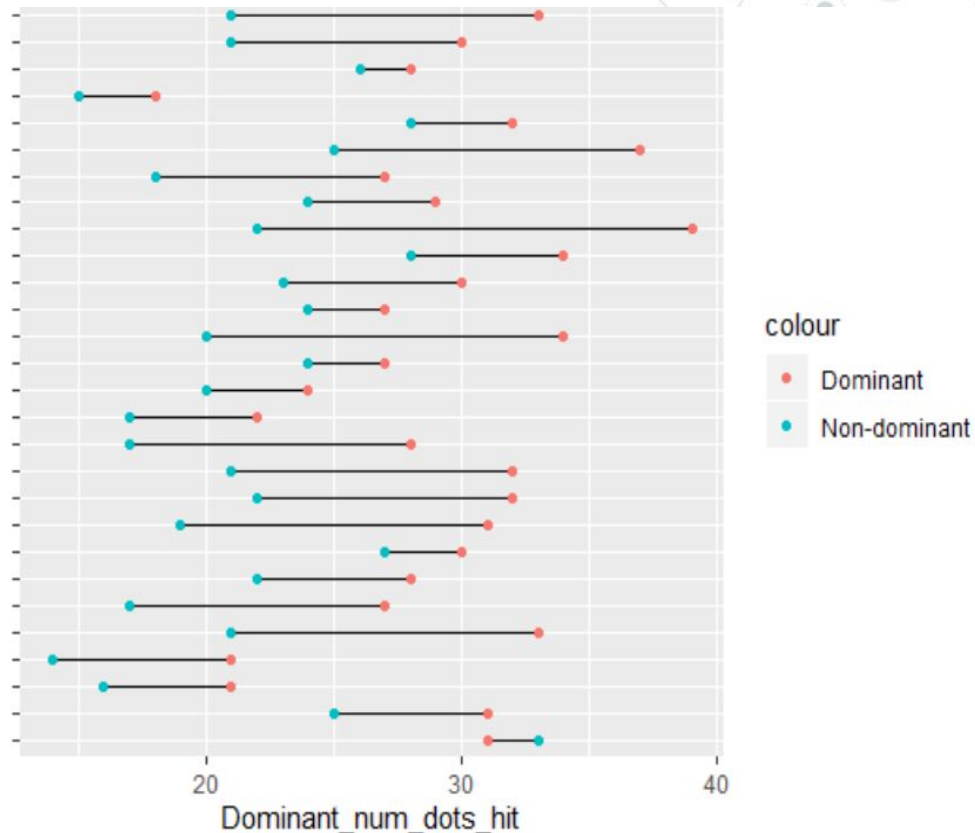
$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2-1}\left(\frac{s_2^2}{n_2}\right)^2}$$

- Test Statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Paired t-test

- Goal: Test if there is a difference in means for the *same* individual at *different* points in time
- Conditions to meet:
 - The data is matched by design (two observations for the each individual)
 - Distribution of difference is normal
- Remember our “Hit the Dot” lab (lab 8)
 - Two points of data came from each student.



Paired t-test cont.

$$H_0 : \mu_d = 0$$

$$H_A : \mu_d \neq 0, \text{ or } \mu_d < 0, \text{ or } \mu_d > 0$$

$$t = \frac{\bar{x}_d - \mu_d}{s_d / \sqrt{n}} \sim t_{n-1}$$



ANOVA and Tukey HSD

ANOVA

⊙ Hypotheses

- **Null hypothesis:**
 - ⊙ $H_0: \mu_1 = \mu_2 = \dots = \mu_K$ where K is the number of levels of the categorical variable
 - ⊙ “There is no difference in the means of the K levels”
(always do this in context)
- **Alternative hypothesis:**
 - ⊙ H_A : Not all $\mu_1, \mu_2, \dots, \mu_K$ are equal
 - ⊙ “At least one of the means differs from the other”

ANOVA cont.

- ◎ The test statistic: ANOVA F statistic follows an F -distribution with $k-1$ dof in the numerator and $N_{total}-k$ dof freedom in the denominator

$$F = \frac{\text{variation among group means}}{\text{variation among individuals in the same group}}$$

$$F = \frac{MSG}{MSE} \sim F_{k-1, N_{total}-k}$$

****Do not need to calculate by hand! Know how to do it in R**

ANOVA cont.

- Mean Squares for groups

$$MSG = \frac{n_1(\bar{x}_1 - \bar{x})^2 + n_2(\bar{x}_2 - \bar{x})^2 + \dots + n_k(\bar{x}_k - \bar{x})^2}{k - 1}$$

- Mean Squares for errors

$$MSE = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \dots + (n_k - 1)s_k^2}{N_{Total} - k}$$

- Note:** you will not need to calculate these by hand! Should know how to find them from R output.

ANOVA in R: use aov() and tidy() it up!

◎ aov() stands for “analysis of variance”

```
> library(broom)
```

```
> cancer_anova <- aov(formula = tumor_volume ~ treatment, data = cancer_data)
```

```
> tidy(cancer_anova)
```

```
## # A tibble: 2 X 6
```

## term	df	sumsq	meansq	statistic	p.value
## <chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## treatment	3	8060	2687	6.70	0.00313
## Residuals	18	7218	401	NA	NA

How do we interpret this output?

Tukey's HSD

Sets overall error rate to be 5% - fixes the issues with multiple testing

Reports the confidence interval and adjusted p-value for each comparison

```
model <- aov(outcome~groups, data = dataset)
tukey <- TukeyHSD(model)
tidy(tukey)
```

A tibble 6X6

## term	comparison	estimate	conf.low	conf.high	adj.p.value
## <chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1 treatment	Cannabinoids-Both	20	13.5	26.5	0.378
## 2 treatment	Irradiation-Both	48.3	41.4	55.1	0.0075
## 3 treatment	Neither-Both	42.3	36.4	48.2	0.0066
## 4 treatment	Irradiation-Cannabinoids	28.2	21.1	35.4	0.190
## 5 treatment	Neither-Cannabinoids	22.3	16.0	28.5	0.263
## 6 treatment	Neither-Irradiation	-5.96	-12.6	0.720	0.964



Non-parametric tests

Non-parametric tests

- © If our data does not meet the assumptions to run a parametric test, we can turn to non-parametric tests
 - These rely on the ranks of the data rather than the numbers themselves.
- © Make sure to understand the steps of running the first two by hand, and what R code to use for all three.
 - Use the lecture slides to see in-depth examples!

Wilcoxon Rank Sum

- ◎ This is the non-parametric counterpart of the two-sample t-test
- ◎ Assumes that the distributions have the same general shape but assumes nothing about the shape
- ◎ Evaluates the null hypothesis that the two populations are identical

Wilcoxon Sign-Rank

- ◎ This is the non-parametric counterpart of the paired t-test
- ◎ Assumes that the distributions have the same general shape but assumes nothing about the shape
- ◎ Evaluates the null hypothesis that the difference between the first and second measures is 0

Kruskal-Wallis

- ◎ This is the non-parametric alternative to the ANOVA test
- ◎ Looks at the medians of each group and determines if they are significantly different
- ◎ Evaluates the null hypothesis that there is no difference between the group medians



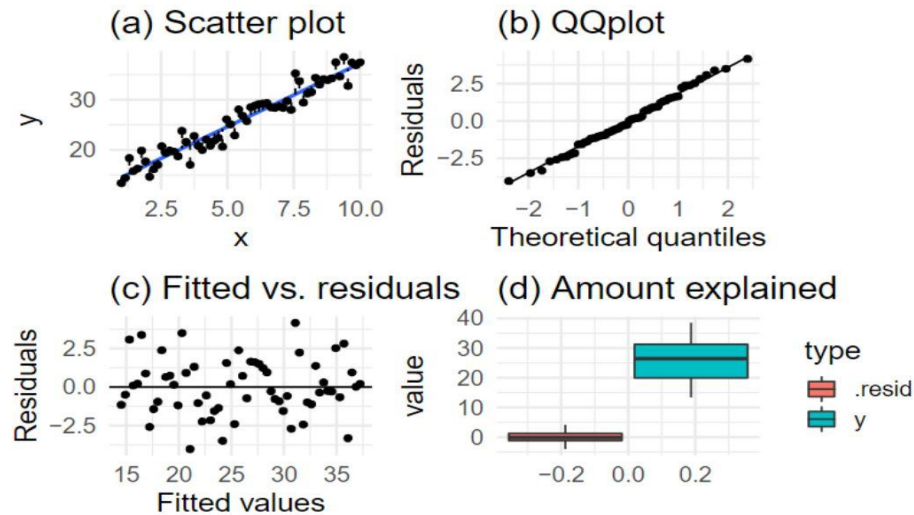
Regression

Regression Inference

- ◎ Assumptions that require checking for regression inference:
 1. The relationship between x and y is linear in the population.
 2. y varies normally around the line of best fit. That is, the **residuals** vary normally around the line of best fit.
 3. Observations are independent.
 4. The standard deviation of the responses is the same for all values of x .
- ◎ Except for assumption #3, these assumptions can be investigated by examining the **estimated residuals**.

So, we make plots about the estimated residuals to check these assumptions.

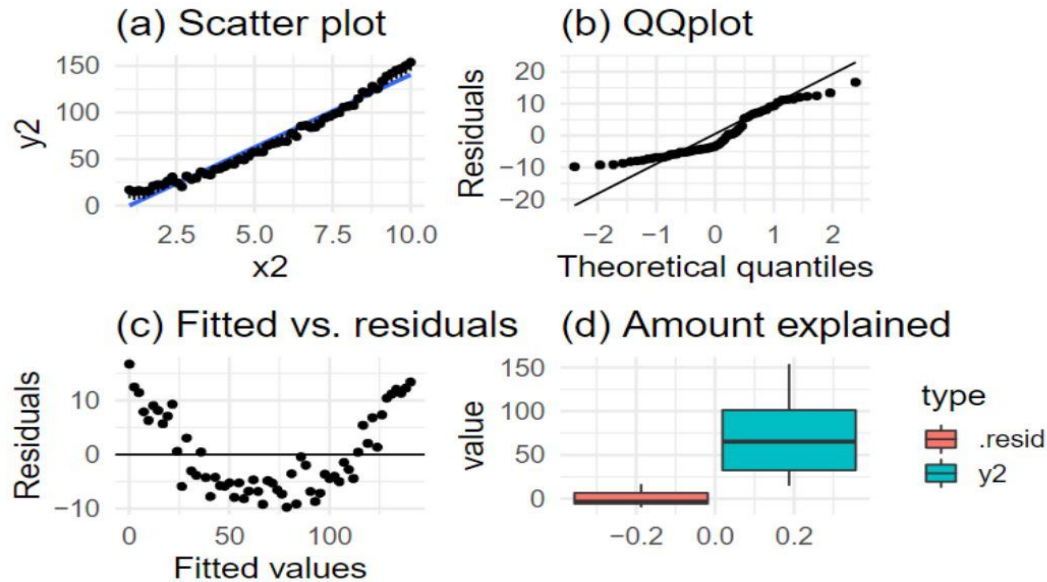
Example 1: Investigating the assumptions



A good fit to the data

- Plot (a) shows a fitted regression line and the data. The estimated residuals are shown by the dashed lines. We want to see that the residuals are sometimes positive and sometimes negative with no trend in their location.
- Plot (b) shows a QQ plot of the residuals (to check if they're normally distributed)
- Plot (c) shows a plot of the fitted values vs. the residuals. We want this to look like a random scatter. If there is a pattern then an assumption has been violated.
- Plot (d) shows a boxplot of the distribution of y vs. the distribution of the residuals. If x does a good job describing y, then the box plot for the residuals will be much shorter because the model fit is good.

Example 2: Investigating the assumptions



The linear relationship assumption does not hold

- Plot (a): While the residuals are small there is a pattern: they start positive, then turn negative and become positive again as x increases.
- Plot (b): The QQ plot does not support normality because it is much different from a line.
- Plot (c): There is a trend in the residuals vs. fitted. This accentuates the pattern observed in plot (a).
- Plots (a)-(c) all provide evidence against the assumption that a linear fit is the most appropriate one. Because the fit is curved, this relationship would require a quadratic function.
- Plot (d): However, even though the linearity assumption is violated, the linear model still explains a lot of the variation so it still offers insight into explaining y , even if it isn't the best model.

Hypothesis Testing for Regression

$H_0: b = 0$ (i.e., There is no association between age and number of cigarettes smoked per day.)

$H_a: b \neq 0$ (i.e., There is an association between age and number of cigarettes smoked per day.)

To test the null hypothesis, the t-test statistic is: $t = \frac{\hat{b}}{SE_b}$

where $SE_b = \frac{s}{\sqrt{\sum(x - \bar{x})^2}}$ and $s = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y - \hat{y})^2}$

We will use R to compute the test statistic, SE_b and s . Be sure you know where SE_b , s , and $\hat{b}$ can be found using the R output and which functions to use to find them.

Confidence Intervals for the regression coefficient

We can also use the output from `tidy(your_lm)` to create a 95% confidence interval for the slope coefficient.

estimate +/- margin of error

$$\hat{b} \pm t^* SE_b$$

Where t^* is the critical value for the t distribution with $n - 2$ degrees of freedom with area C (e.g., 95%) between $-t^*$ and t^* .

Inference for prediction of average vs. individual response

- ◎ If you want to make inference for the **mean response** μ_y when x takes the value x^* :

$$\hat{y} \pm t * SE_{\hat{\mu}}, \text{ where } SE_{\hat{\mu}} = s \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x - \bar{x})^2}}$$

- ◎ If you want to make inference for a **single observation** y when x takes the value x^* :

$$\hat{y} \pm t * SE_{\hat{y}}, \text{ where } SE_{\hat{y}} = s \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x - \bar{x})^2}}$$

As a summary...

If you have continuous data

